

$$1. \quad P(k, k+1) = a \frac{4-k}{4} \quad P(k, k-1) = b \frac{k}{4} \quad P(k, k) = 1 - P(k, k+1) - P(k, k-1)$$

a) Detailed Balance condition

$$\pi(x) P(x, y) = \pi(y) P(y, x)$$

$$P(k, k) = 1 - P(k, k+1) - P(k, k-1)$$

$$\theta = \frac{a}{a+b} \quad 1+\theta = \frac{b}{a+b}$$

$$P(k, k-1) = b \frac{k}{4} \Rightarrow P(k+1, k) = b \frac{k+1}{4}$$

$$\pi(k) P(k, k+1) = \pi(k+1) P(k+1, k)$$

$$\Rightarrow \pi(k) a \frac{4-k}{4} = \pi(k+1) b \frac{k+1}{4}$$

$$\Rightarrow \pi(k+1) = \pi(k) \frac{a}{b} \frac{4-k}{k+1} = \left(\frac{a}{b}\right)^2 \frac{5-k}{k} \frac{4-k}{k+1} \pi(k-1) \cdot \frac{a}{b} \frac{k+1}{k}$$

$$\pi(k) P(k, k-1) = \pi(k-1) P(k-1, k) = \pi(k) b \frac{k}{4} = \pi(k-1) a \frac{5-k}{4} \Rightarrow \pi(k) = \frac{b}{a} \frac{k}{5-k} \pi(k-1)$$

$$\therefore \pi(k+1) = \left(\frac{a}{b}\right)^2 \frac{b}{a} \frac{4-k}{k+1} \frac{k}{5-k} \pi(k) = 1 \cdot \frac{4-k}{k+1} \frac{k}{5-k} \dots$$

$$\Rightarrow \pi(2) = \frac{a}{b} \left(\frac{4-1}{2}\right) \pi(1) = \frac{a}{b} \left(\frac{3}{2}\right) \cdot \frac{a}{b} \left(\frac{4}{1}\right) \pi(0) = \left(\frac{a}{b}\right)^2 \left(\frac{4!}{2!}\right) \pi(0) = \left(\frac{a}{b}\right)^2 \left(\frac{4!}{2!}\right)$$

$$\sum_{i=1}^{k+1} \pi(i) = 1 \Rightarrow \pi(k) \frac{a}{b} \frac{4-k}{k+1} + \pi(k-1) \frac{5-k}{k \cdot b} + \dots + \frac{a}{b} \left(\frac{2}{3}\right) \pi(1) + \pi(1) = 1$$

$$\Rightarrow \pi(1) \left[\left(\frac{a}{b}\right)^{k+1} \frac{k+1!}{4} + \dots + \pi(k) \right] = 1$$

$$\pi(4) = \left(\frac{a}{b}\right)^4 \frac{1}{4} \pi(3) = \frac{a}{b} \cdot \frac{1}{4} \cdot \frac{a}{b} \cdot \frac{2}{3} \cdot \frac{a}{b} \cdot \frac{3}{2} \cdot \frac{a}{b} \cdot \frac{4}{1} = \left(\frac{a}{b}\right)^4$$

$$\pi(3) = \frac{a}{b} \frac{2}{3} \pi(2)$$

$$\pi(k) = \left(\frac{a}{b}\right)^k$$

$$\pi(2) = \frac{a}{b} \frac{3}{2} \pi(1)$$

$$\pi(1) = \frac{a}{b} \frac{4}{1}$$

1. b) $a=6=1$ $q_0 = \int_0^1$ $q_n = q_0 p^n$ $n \leq 60$

$$q_{50} = \begin{bmatrix} .3863 \\ -1.1442 \\ .9977 \\ -.02204 \\ 0 \end{bmatrix} \times 10^8 \quad q_{51} = \begin{bmatrix} -5721 \\ 1.6922 \\ -1.4445 \\ .3259 \\ 0 \end{bmatrix} \times 10^8$$

q_n fails to converge b/c odd powers cause sign change

While π is unique b/c π does not determine q

$p^n(k, k) = 0$ for odd n since $p(k, k) = 1 - \underbrace{p(k, k+1) - p(k, k-1)}_{\text{sum to 1}} \geq 1 - 1 = 0$

c) $p_1 = \begin{bmatrix} .7 & .3 & 0 & 0 & 0 \\ .05 & .65 & .3 & 0 & 0 \\ 0 & .05 & .325 & .1 & 0 \\ 0 & 0 & .1 & .9 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

for $n=1$

$$\pi = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

analytical

$\pi = \text{binomial}(4, 1) \quad \theta = \frac{.3}{.3+.1} = \frac{3}{4}$

$k=0, 1, 2, 3, 4$ $= \frac{4!}{k!(4-k)!} (3/4)^k (1/4)^{4-k}$

$\Rightarrow \pi = \begin{bmatrix} .004 \\ .0469 \\ .2109 \\ .4219 \\ .3964 \end{bmatrix}$

could not get $\max_k [q_n(k) - \pi(k)] < 10^{-6}$
to converge

$$2. a) A_{ij} = 1 - A_{ji}$$

$$j \neq i \quad P(i, i) = 1 - \sum_{j \neq i} P(i, j)$$

$$P(i, j) = A_{ji} / (n-1)$$

$$A_{ij} = \frac{s_i}{s_i + s_j}$$

$$\pi(i) P(i, j) = \pi(j) P(j, i) \quad P(j, i) = \frac{A_{ji}}{n-1} \quad P(i, j) = \frac{A_{ji}}{n-1} = \frac{1 - A_{ij}}{n-1}$$

$$\Rightarrow \pi(i) = \frac{A_{ij}}{n-1} \cdot \frac{n-1}{1 - A_{ij}} \pi(j) \Rightarrow \pi(i) = \frac{A_{ij}}{1 - A_{ij}} \pi(j) \quad \text{not dependent on } n$$

$$n=3 \quad S = \{3, 2, 1\}$$

$$P = \begin{bmatrix} .325 & .3 & .375 \\ .2 & .467 & .333 \\ .125 & .167 & .703 \end{bmatrix}$$

$$A_{1, S} = \begin{bmatrix} X & \frac{3}{3+2} & \frac{3}{3+1} \end{bmatrix} = \begin{bmatrix} X & .6 & .75 \end{bmatrix}$$

$$A_{2, S} = \begin{bmatrix} \frac{2}{2+3} & X & \frac{2}{2+1} \end{bmatrix} = \begin{bmatrix} .4 & X & .67 \end{bmatrix}$$

$$A_{3, S} = \begin{bmatrix} \frac{1}{1+3} & \frac{1}{1+2} & Z \end{bmatrix} = \begin{bmatrix} .25 & .33 & Z \end{bmatrix}$$

$$P = \begin{bmatrix} X = \frac{.6}{2} & \frac{.75}{2} \\ \frac{.4}{2} & Y & \frac{.67}{2} \\ \frac{.25}{2} & \frac{.33}{2} & Z \end{bmatrix} = \begin{bmatrix} X & .3 & .375 \\ .2 & Y & .333 \\ .125 & .167 & Z \end{bmatrix}$$

weakest player is 3

$$\Rightarrow \text{mean return time} = E_3 T_3$$

$$E_3 T_3 = \frac{1}{\pi_3}$$

$$\pi(3) = \frac{s_3}{s_1 + s_2 + s_3} = \frac{1}{3+2+1} = \frac{1}{6}$$

$$E_3 T_3 = \frac{1}{\frac{1}{6}} = 6 \quad \boxed{E_3 T_3 = 6}$$

$$X = 1 - .3 - .375 = .325$$

$$Y = 1 - .2 - .333 = .467$$

$$Z = 1 - .125 - .167 = .703$$

$$2. b) \pi_1 = \frac{s_1}{s_1+s_2+s_3} \quad \pi_2 = \frac{s_2}{s_1+s_2+s_3} \quad \pi_3 = \frac{s_3}{s_1+s_2+s_3}$$

$$\Rightarrow \pi_1 P(1,2) \pi_2 P(2,3) \pi_3 P(3,1) = \pi_1 P(1,3) \pi_3 P(3,2) \pi_2 P(2,1)$$

$$P(1,2)P(2,3)P(3,1) = P(1,3)P(3,2)P(2,1)$$

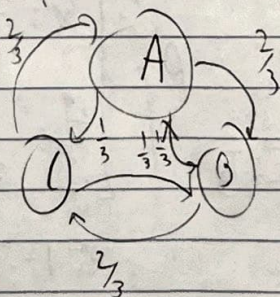
$$A_{AB} = \frac{2}{3} \quad A_{BC} = \frac{2}{3} \quad A_{CA} = \frac{2}{3}$$

$$A_{AC} = A_{CB} = A_{BA} = 1 - A_{AB} = \frac{1}{3}$$

$$1 - \frac{1}{3} - \frac{2}{3} = 0 \Rightarrow A_{AA} = A_{BB} = A_{CC} = 0$$

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}$$

for all columns and rows
 $\frac{2}{3} + \frac{1}{3} = 1 \Rightarrow$ doubly stochastic



$\Rightarrow$ images equal strength

$$\Rightarrow \pi_1 = \pi_2 = \pi_3$$

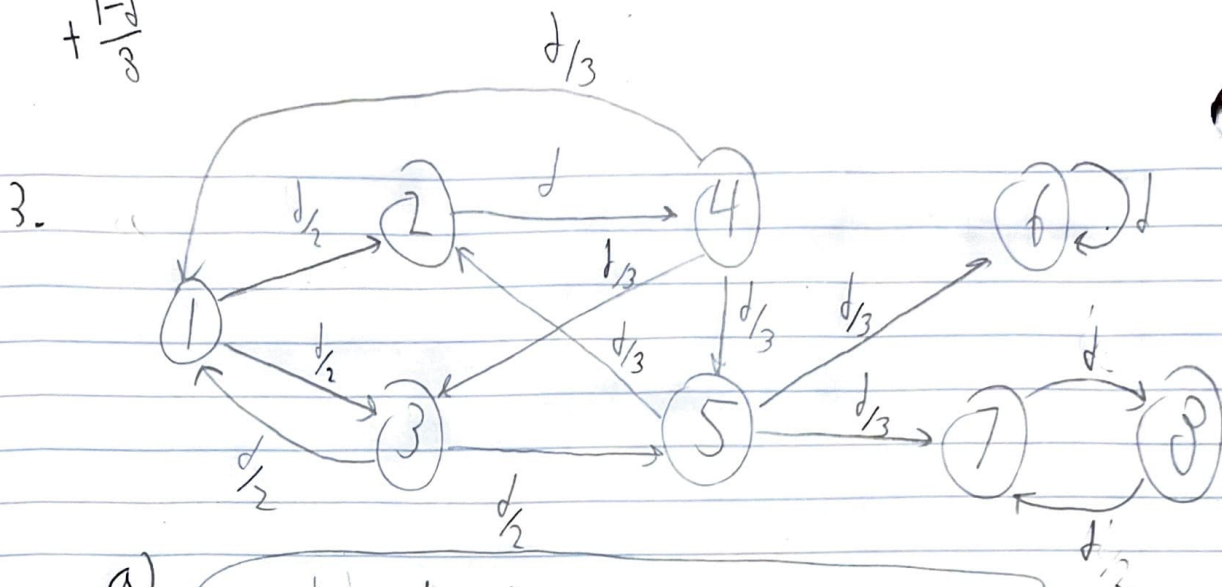
$$\pi_A P(A,B) = \pi_B P(B,A)$$

$$\pi_A = \pi_B$$

$\Rightarrow P(A,B) = P(B,A)$ False statement

$$\pi_i P(i,j) - \pi_j P(j,i) = 0 \text{ from diagram}$$

$$+ \frac{1}{2}$$



a)

$P =$

0	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	0	0
0	0	0	1	0	0	0	0
$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	0	0	0
$\frac{1}{3}$	0	$\frac{1}{3}$	0	$\frac{1}{3}$	0	0	0
0	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0
0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0

States 6 and $\{7, 8\}$ are closed since there are no outgoing communicating arrows

using MATLAB

$$q_{100} = q_0 \rho^{100} =$$

$\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \right]$
 $\bullet 4375$
 $\bullet 2695$
 $\bullet 2930$

$$q_{|0|} = q p^{|0|} =$$

$\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ .4875 \\ .2695 \\ .2930 \end{array} \right]$

ρ falls to

$$3.1) G = \frac{1}{2} P + \frac{1-\beta}{2} \mathbf{1}\mathbf{1}^T \quad \beta = 2.85$$

$$\Pi G = \Pi$$

$$q_{n+1} = q_n G$$

$$\Rightarrow q_{n+1} - \Pi = q_n G - \Pi$$

$$\Rightarrow q_{n+1} - \Pi = q_n G - \Pi G$$

$$\stackrel{1}{=} (q_n - \Pi) G$$

$$= (q_n - \Pi) \left(\frac{1}{2} P + \frac{1-\beta}{2} \mathbf{1}\mathbf{1}^T \right)$$

$$= \frac{1}{2} (q_n - \Pi) P + \frac{1-\beta}{2} (q_n - \Pi) \mathbf{1}\mathbf{1}^T \Rightarrow q_n - \Pi = \frac{1}{2} (q_n - \Pi) P$$

if take 4 iterations

Pages 1 and 5 tie b/c they have the same return probabilities

$$\|q_n - \Pi\|_1 \leq \beta^n \|q_0 - \Pi\|_1$$

$$\| \frac{1}{2} (q_{n-1} - \Pi) P \|_1 \leq \beta^n \|q_0 - \Pi\|_1$$

$\downarrow$

$$\|VP\|_1 \leq \|V\|_1$$

makes Π unique since only 1 value V works